LOPERAMIDE

Not in bloody diarrhoea

Cardiac toxicity in overdose

CHEMICAL NAME 4-[4-(4-chlorophenyl)-4-hydroxypiperidin-1-yl]-N,N-dimethyl-2,2-diphenylbutanamide (hydrochloride)

CLASS Peripheral opioid-receptor agonist antidiarrhoeal

TRADE NAMES Imodium · Diarrol · Lopamide · Imodium Instants

THERAPEUTIC USES / INDICATIONS

- Acute non-infective or mild travellers' diarrhoea (adults).
- Chronic diarrhoea (IBS-D, after bowel resection, high-output stoma).

ADMINISTRATION

- **PO** capsules, tablets or orodispersible tablets after each loose stool.
- Always with oral rehydration.

DOSAGES

- **Acute diarrhoea (adult):** 4 mg initially, then 2 mg after each loose stool; max 16 mg/day for up to 5 days (OTC max 12 mg/48 h).
- **Chronic diarrhoea:** 4–8 mg/day in divided doses.
- **Children 4–8 yr:** 1 mg 3–4 times daily (specialist; not <4 yr in most guidance).

PHARMACOKINETICS

- **Onset:** 1 h. **Duration:** up to 24 h.
- Extensive first-pass metabolism (CYP3A4, 2C8); P-gp keeps it out of brain.
- Excreted in faeces. **t½:** 9–14 h.

PHARMACODYNAMICS

- Agonist at **μ-opioid receptors** in gut wall → ↓ peristalsis, ↑ transit time, ↑ anal sphincter tone, ↓ secretion.
- No CNS effects at normal doses.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Constipation.
- Abdominal cramps, bloating.
- Nausea, dizziness.

ADVERSE EFFECTS (SERIOUS)

- Paralytic ileus, toxic megacolon (in colitis or infective dysentery).
- QT prolongation, torsades, cardiac arrest with high doses/misuse.
- CNS depression in young children.

CONTRAINDICATIONS & CAUTIONS

- Bloody diarrhoea or high fever (dysentery: Shigella, Salmonella, *C. difficile*, E. coli O157).
- Acute ulcerative colitis, pseudomembranous colitis.
- Children <4 yr (≤2 yr absolutely).
- Where constipation, ileus or abdominal distension must be avoided.

KEY INTERACTIONS

- **P-gp/CYP3A4 inhibitors** (ritonavir, itraconazole, quinidine): ↑ levels, CNS effects.
- **QT drugs:** caution.

NURSING MANAGEMENT

- Check stools for blood, temperature, recent antibiotic use (C. diff) before giving.
- Make sure rehydration.
- Stop if abdominal distension, constipation or ileus.
- Monitor stool frequency.

PATIENT EDUCATION

- Drink plenty of fluids (ORS).
- Stop and seek help if blood in stools, fever, swollen abdomen or no improvement in 48 h.
- Never exceed the maximum dose.
- Not for young children.

KEY POINTS

- Peripheral μ-agonist antidiarrhoeal.
- Never in bloody/infective dysentery or C. diff.
- 4 mg then 2 mg per loose stool, max 16 mg/day.
- Overdose → cardiac arrhythmias.

SOURCES CHECKED **[566]** Loperamide 2 mg Capsules (UK SmPC) **[567]** Loperamide Hydrochloride, oral (US label) **[568]** Loperamide Hydrochloride, oral (US label, Bryant Ranch Prepack)

Full references p. 432–457